Detailed Trading Requirements Spec

Part 2: Institutional System B, Exits, Circuits & Scanner

5. System B: Institutional Edge Indicators

System B measures market microstructure. Rather than observing price lags, System B observes buying/selling pressures before they reflect in price changes. It parses order flow feeds directly from IBKR tick data.

1. Volume Delta & Cumulative Delta

Volume Delta is computed by comparing tick trade prices against the prevailing bid/ask mid-price. Trades executing closer to the ask are categorized as aggressive buys; trades executing closer to the bid are aggressive sells. Cumulative Delta sums these delta readings over the session to show who maintains net inventory control.

2. DOM Imbalance

Formula: Imbalance = (Sum of Bid Sizes in Top 10 Levels) / (Sum of Ask Sizes in Top 10 Levels).
Why: High imbalance indicates a depth-of-book support wall. If bids stack at 3x ask depth, it represents institutional buying interest waiting to absorb market orders.

3. Volume Profile & POC (Point of Control)

Volume Profile plots trading volume at specific price increments. The single price level that accumulated the highest volume during the session is the POC. Value Area High (VAH) and Value Area Low (VAL) bound the range containing 70% of the session's volume. Price pulling back to a high-volume node like POC is a high-probability bounce entry.

4. Options Chain Activity

The Options Scanner monitors: (a) Put/Call volume ratios to track macro hedging; (b) Strike concentrations to find 'Max Pain' (where market makers benefit most from expiration pinning); and (c) Unusual options flow, flagging strikes where volume exceeds 2.5x open interest.

5. Dark Pool Block Approximation

Since dark pool trades are off-exchange block crosses, they print on the tape with distinct characteristics. The system flags trades of > 10,000 shares executing exactly at the bid/ask mid-price. Accumulation patterns are flagged when multiple block crosses print without causing immediate price fluctuations.

6. Detailed Exit Scenarios & Trailing Stop Math

Exit signals are strictly managed via a position tracker state machine. To maximize gains and limit losses, exit rules are partitioned by signal classification.

1. Trend Continuation Positions (2-of-5 Rule)

A Trend position is held for larger moves. It exits only if 2 of the following 5 conditions trigger simultaneously:
- MACD histogram decline: Histogram falls for 3 consecutive 5m candles.
- Volume Climax: Candle volume is > 2.5x average AND price closes in the bottom 30% of its high-low range (exhaustion signature).
- RSI momentum break: RSI falls back below 50 after crossing above 60.
- EMA structure break: Price closes below the 9 EMA for 2 consecutive 5m bars.
- Trailing Stop-Loss: The locked trailing stop level is breached.

2. Momentum Scalp Positions (1-of-4 Rule)

A Scalp position targets quick in-out moves. It exits immediately if any 1 of these 4 conditions is met:
- MACD histogram decline: Histogram falls for 2 consecutive candles.
- Volume exhaustion: Candle volume falls to < 50% of the 20-bar average.
- Profit Target reached: Price hits Target 1 (calculates auto-exit).
- Trailing Stop-Loss: Tighter trailing stop level (1.0x ATR multiplier) is breached.

3. Trailing Stop-Loss Mathematical Formula

On position entry:
  Stop_0 = Entry_Price - (ATR(14) * Multiplier)

On every new 5m candle close:
  Candidate_Stop_t = Close_t - (ATR(14) * Multiplier)
  Stop_t = Max(Stop_{t-1}, Candidate_Stop_t)

This creates an upward-only ratchet. The stop level can never decline, locking in profits during rallies.

7. Risk Control & SEC Filing Agent

Multi-Level Circuit Breakers

To prevent runaway losses on high-correlation days, the CEO agent tracks daily wins/losses:
- Level 1: 2 consecutive losses. Triggers capital-preservation warnings.
- Level 2: 3 consecutive losses. Reduces all signal scores by 20 points. Disables all Momentum Scalp signals (only Dual Confirm allowed).
- Level 3: 4 consecutive losses OR daily drawdown > 2.0%. Halts the trading system entirely for the rest of the session.

SEC EDGAR filing Scanner

The SEC agent queries the SEC EDGAR EFTS JSON index API to track smart money actions:
- Monitors Form 4 filings for insider transactions. Flags transactions where corporate insiders buy > $1M of stock.
- Monitors SC 13D filings for activist accumulations of > 5% of outstanding shares.
- Monitors 8-K filings for merger and acquisition activity.
- Schedule: Scans every 10 minutes during market sessions, and every 15 minutes in after-hours (until 11:00 PM ET). Runs weekend sweeps at 10:00 AM on Saturdays and Sundays.

Reports & UI Functions

The system issues regular performance reports via Telegram:
- Daily Close (4:05 PM ET): Summarizes trades, P/L, win rates, and maps tomorrow's economic calendar and stock support levels.
- Weekly Homework (Friday 4:30 PM ET): Evaluates indicator performance and suggests automated score threshold updates.
- Streamlit UI: Searchable US stock list, threshold controls, active position states, and a Backtest Laboratory with Signal Replay.